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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Smart EV Charging Station Queue and Slot Scheduling System

CSA0604 - DESIGN AND ANALYSIS OF ALGORITHMS

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Introduction

- The **Smart EV Charging Station Queue and Slot Scheduling System** is designed to manage the increasing demand for electric vehicle (EV) charging stations.
- As the number of EVs increases, charging stations can experience **long queues, limited charging slots, and increased waiting time**.
- Traditional **First-Come-First-Served (FCFS)** systems may not efficiently handle different charging requirements, battery levels, and urgent users.
- The proposed system allows EV users to **book charging slots in advance** and receive suitable charging schedules.

Problem Statement

- Existing charging stations experience long waiting times during peak hours.
- Manual slot allocation makes the charging process inefficient.
- Users cannot easily know the real-time availability of charging slots.
- Poor queue management results in wasted time and energy.
- An intelligent scheduling system is required to improve charging efficiency.

Objectives

- To manage EV users, vehicles, charging stations, and slots efficiently.
- To provide secure registration and easy vehicle information management.
- To schedule charging slots and manage vehicle queues effectively.
- To monitor charging activities and generate useful system reports.

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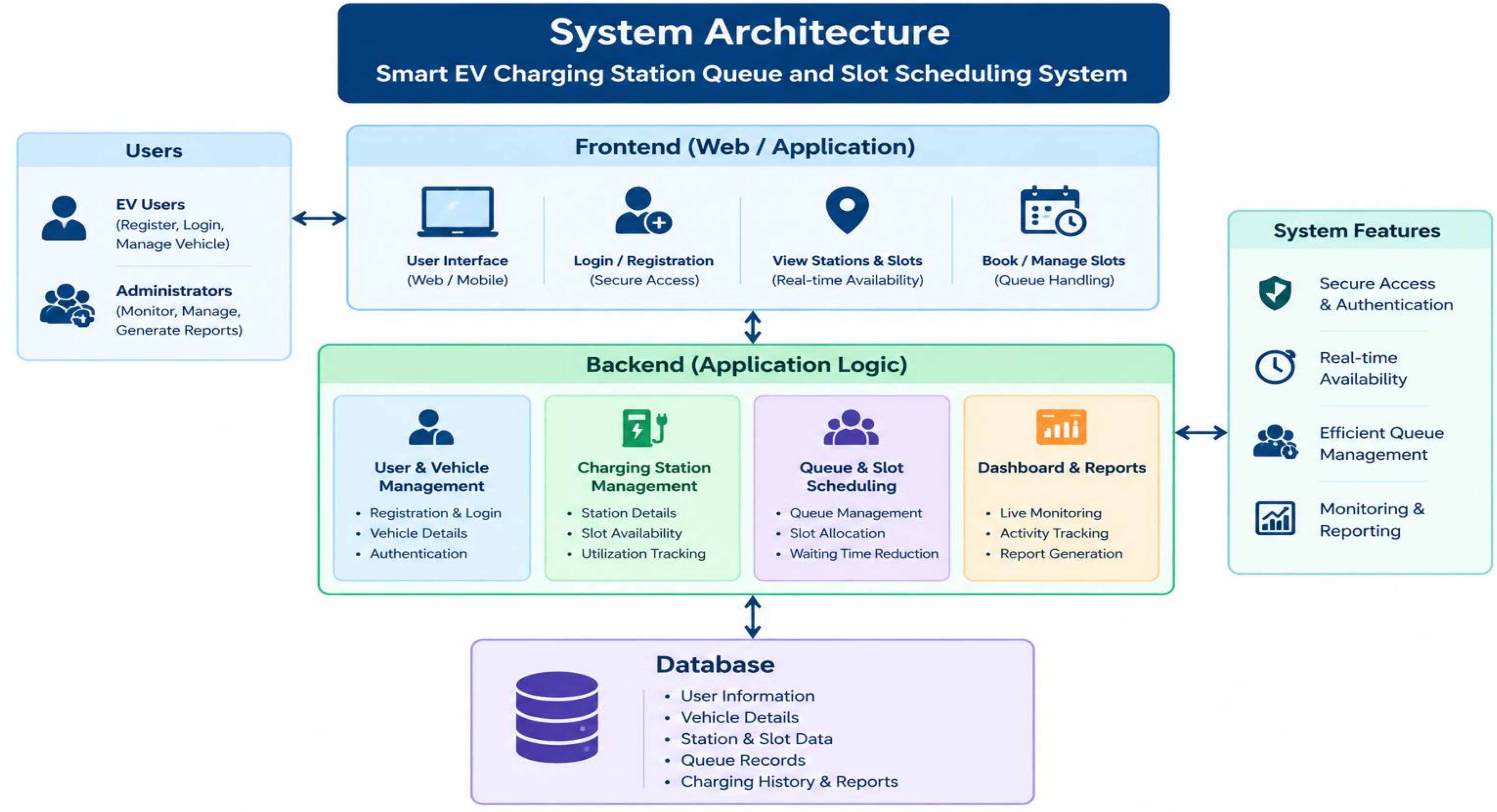
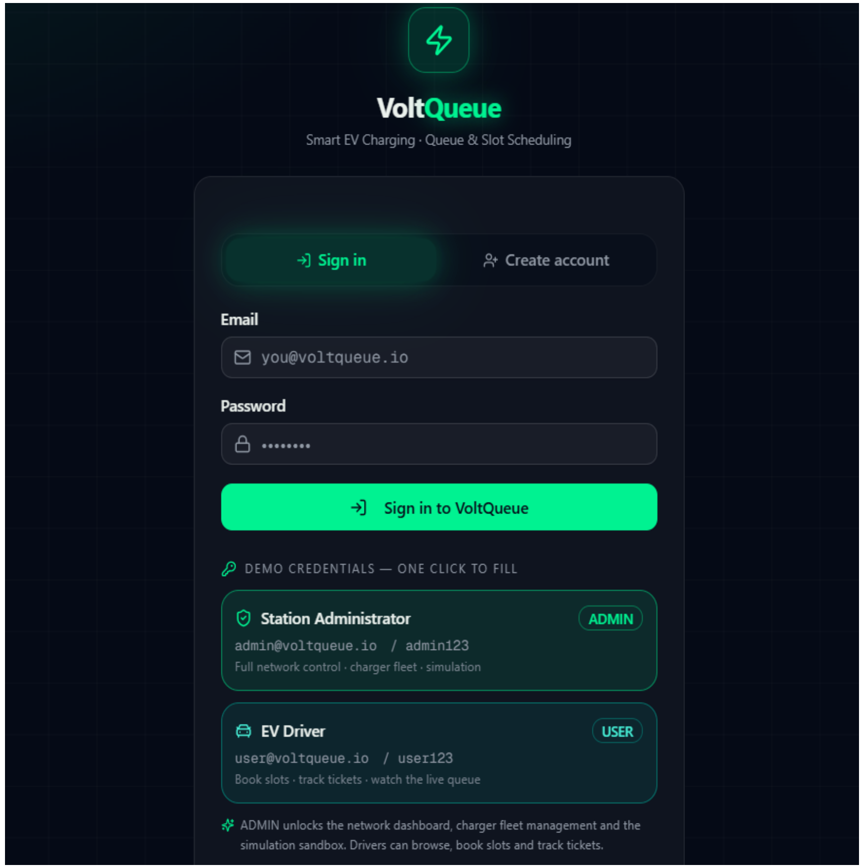
Literature Review

Author / Year	Research Focus	Approach	Key Finding	Gap / Limitation
Wang et al., 2025	EV charging queue management	Queuing model + bi-level scheduling	Improved charging demand fulfillment and reduced queuing time.	Focuses strongly on power constraints and optimization. (MDPI)
Varshney et al., 2025	EV charging station queues	Stochastic modeling + queueing analysis	Models waiting, abandonment, station utilization and throughput.	More analytical; does not provide a complete user-facing management system. (Nature)
Upadhyaya et al., 2026	Fast-charging scheduling	Online scheduling + optimization	Reported reduced waiting time under limited charger availability.	Mainly concentrates on scheduling optimization rather than integrated station management. (Taylor & Francis Online)
Liu & Zhu, 2026	EV charging allocation	Two-stage allocation + congestion-aware queuing	Uses station-level allocation to control congestion and improve charging assignment.	Primarily focuses on network-level allocation. (arXiv)
Kahlert et al., 2026	Charging congestion forecasting	Fluid-queue forecasting	Forecasting congestion can improve charging schedules and reduce waiting-related downtime.	Focuses on forecasting rather than complete queue, slot and dashboard management. (arXiv)


System Architecture
Smart EV Charging Station Queue and Slot Scheduling System
Users
EV Users
(Register, Login, Manage Vehicle)
Administrators
(Monitor, Manage, Generate Reports)
Frontend (Web / Application)
User Interface
(Web / Mobile)
Login / Registration
(Secure Access)
View Stations & Slots
(Real-time Availability)
Book / Manage Slots
(Queue Handling)
Backend (Application Logic)
User & Vehicle Management
• Registration & Login
• Vehicle Details
• Authentication
Charging Station Management
• Station Details
• Slot Availability
• Utilization Tracking
Queue & Slot Scheduling
• Queue Management
• Slot Allocation
• Waiting Time Reduction
Dashboard & Reports
• Live Monitoring
• Activity Tracking
• Report Generation
System Features
Secure Access & Authentication
Real-time Availability
Efficient Queue Management
Monitoring & Reporting
Database
• User Information
• Vehicle Details
• Station & Slot Data
• Queue Records
• Charging History & Reports

Module-1: User Registration and Login Module

- Allows new users to register in the system.
- Stores user details such as name, email, phone number, and EV details.
- Provides secure login using username/email and password.
- Allows users to update their profile information.
- Maintains user account and vehicle information.
- Provides access to booking and charging services after login.



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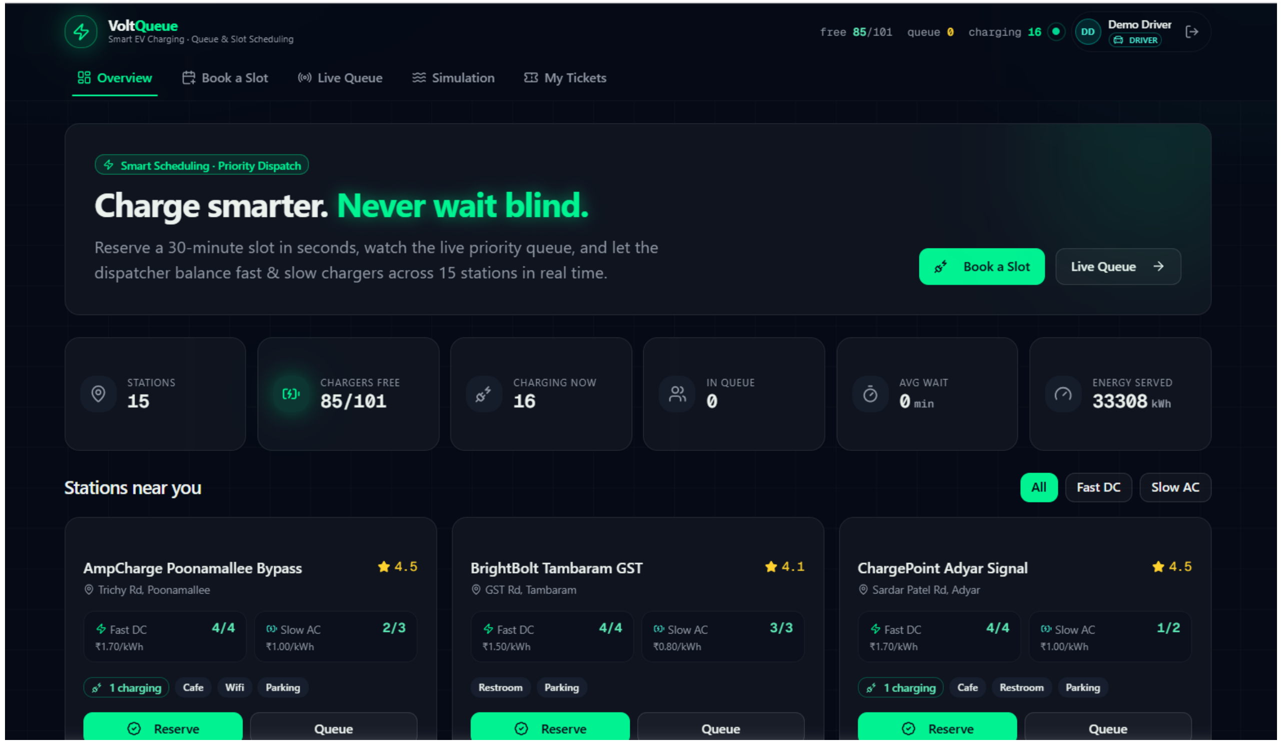
Module 2: Charging Station and Slot Management Module

- Manages available EV charging stations.
- Displays the number of available and occupied charging slots.
- Stores charger details such as charger type and charging capacity.
- Updates slot status in real time.
- Allows administrators to add, update, or remove charging stations.
- Helps users view available charging slots.



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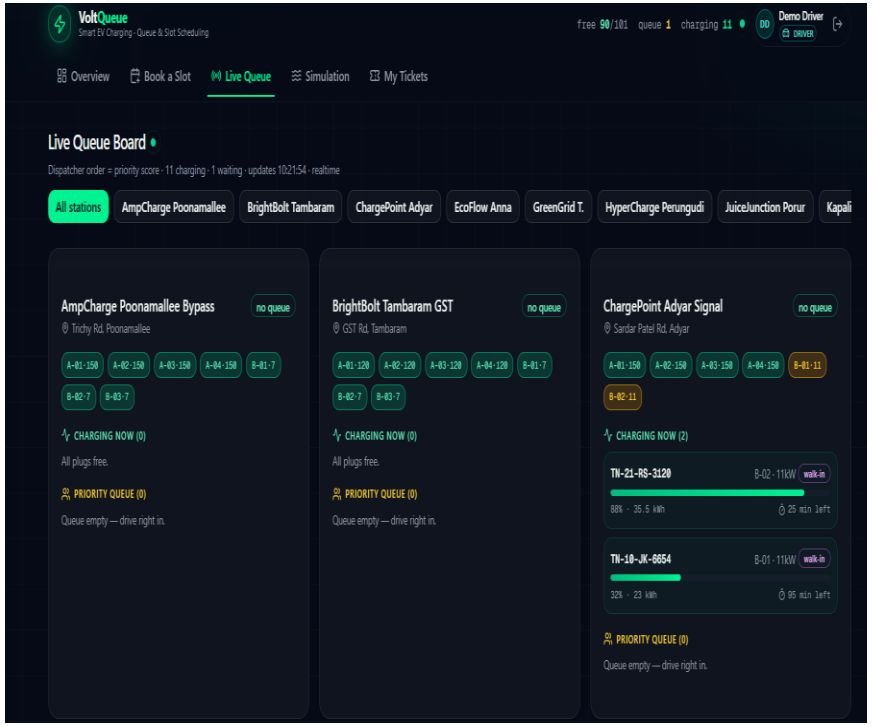
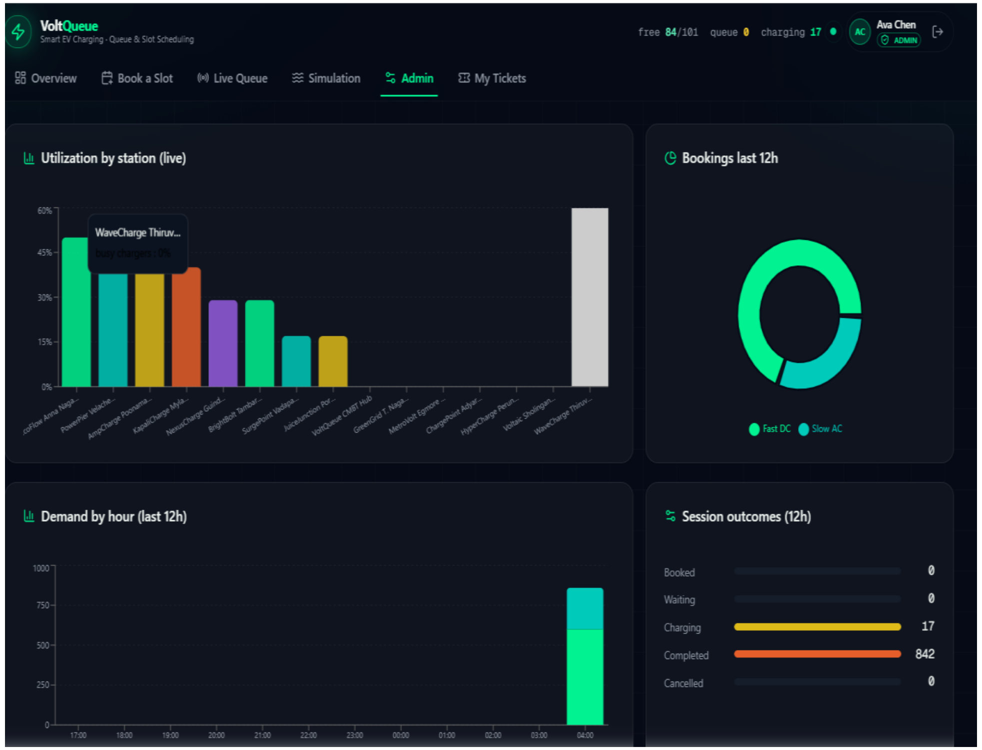
Module 3: Charging Slot Allocation and Queue Management Module

- ➢ Allocates charging slots based on availability.
- ➢ Maintains a queue when all charging slots are occupied.
- ➢ Considers arrival time, battery level, charging duration, and priority.
- ➢ Reduces unnecessary waiting time for EV users.
- ➢ Automatically assigns the next available slot.
- ➢ Supports cancellation and rescheduling of bookings.



Module-4: Charging Monitoring and Report Generation Module

- Monitors ongoing charging activities.
- Records charging start time and completion time.
- Tracks charging duration and energy consumption.
- Displays the current status of each charging session.
- Generates daily, weekly, and monthly reports.
- Provides information about station usage, waiting time, and completed charging sessions.



Tools and Technologies

Component	Technology / Tool Used
AI-Assisted Development	Z.AI
Frontend Development	HTML5, CSS3
Client-Side Scripting	JavaScript
Backend / Server-Side	Web-based Server Environment
Data Management	Database Management System
Deployment	Live Web Server

Algorithm / Technique	Application in the Project
FCFS Scheduling	Organizes charging requests based on arrival order.
Greedy Approach	Selects a suitable charging slot based on current availability.
Dijkstra's Algorithm	Finds the shortest route to a charging station.
Priority Scheduling	Manages charging requests according to priority.
Sequential Search	Searches available charging slots.

Future Scope and Applications

Future Scope

- AI-based prediction of charging demand during peak hours.
- Dynamic slot scheduling based on real-time availability.
- Mobile application for booking and monitoring charging slots.
- Real-time notifications for slot allocation and queue status.
- Smart energy management for efficient charging.

Applications

- EV charging stations and charging hubs.
- Shopping malls and commercial parking areas.
- Colleges and office campuses.
- Smart parking facilities with EV charging.
- Smart city EV infrastructure management.

Conclusion

- The Smart EV Charging Station Queue and Slot Scheduling System provides an organized solution for managing EV charging services.
- It enables users to manage their profiles and vehicle information efficiently.
- Charging stations and available slots can be monitored and managed systematically.
- Queue management and automated slot allocation help reduce waiting time and improve slot utilization.
- The system provides a centralized dashboard and reports for better monitoring and decision-making.

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THANK YOU!